

INTRODUCTION MACHINE LEARNING

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OUTLINE

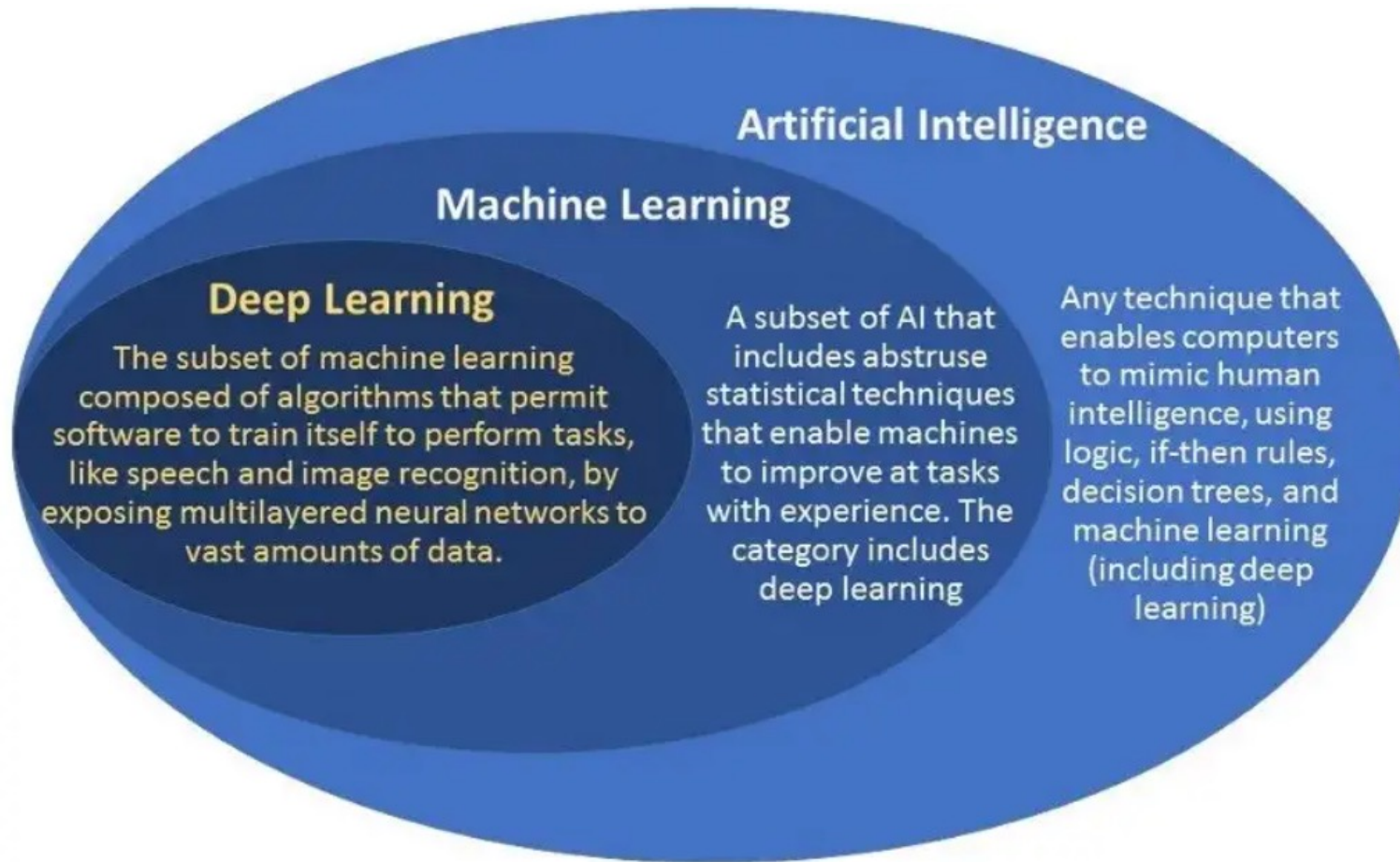
- Overview Machine Learning
- Type of Machine Learning
- Machine Learning Life Cycle
- Applications of Machine Learning in real-world scenarios



OVERVIEW MACHINE LEARNING



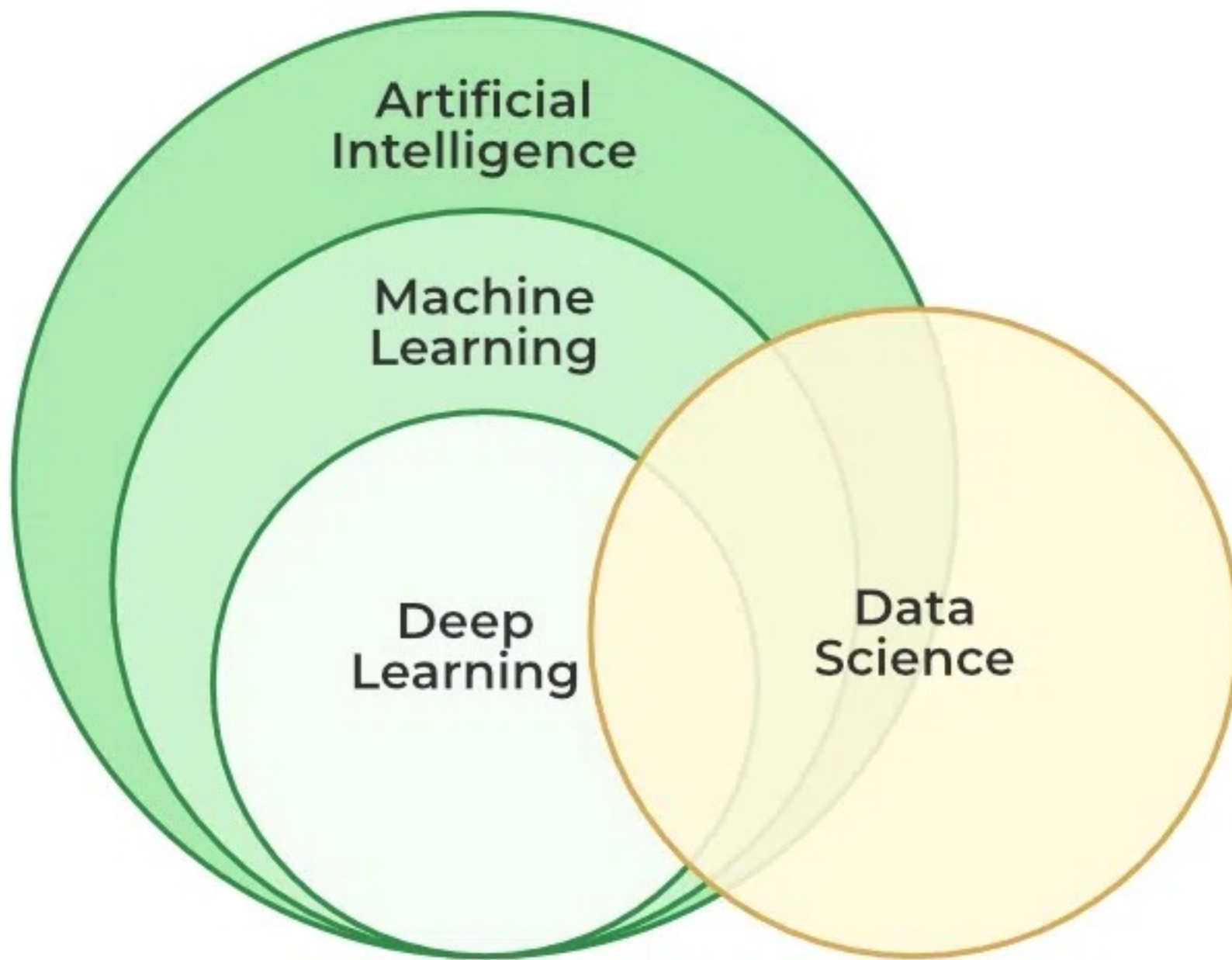
WHAT IS MACHINE LEARNING ?



ML VS DL

Differences	Machine Learning	Deep Learning
Human Supervision	Machine learning requires more supervision.	Deep learning models require almost no human supervision after development.
Hardware Resources	You build and run Machine learning programs on a powerful CPU.	Deep learning models require more powerful hardware, like dedicated GPUs.
Time & Effort	The time required to set up a Machine learning model is less than deep learning, but its functionality is limited.	It requires more time to develop and train data with deep learning. Once created, it continues to improve its accuracy with time.
Data (structured/unstructured)	Machine learning models need structured data to give results (except unsupervised learning) and require continuous human intervention for improvement.	Deep learning models can process unstructured and complex datasets without compromising accuracy.
Use-cases	eCommerce websites and streaming services that use recommendation engines.	High-end applications like Autopilot in planes, self-driving vehicles, Rovers on the Martian surface, face recognition, etc.

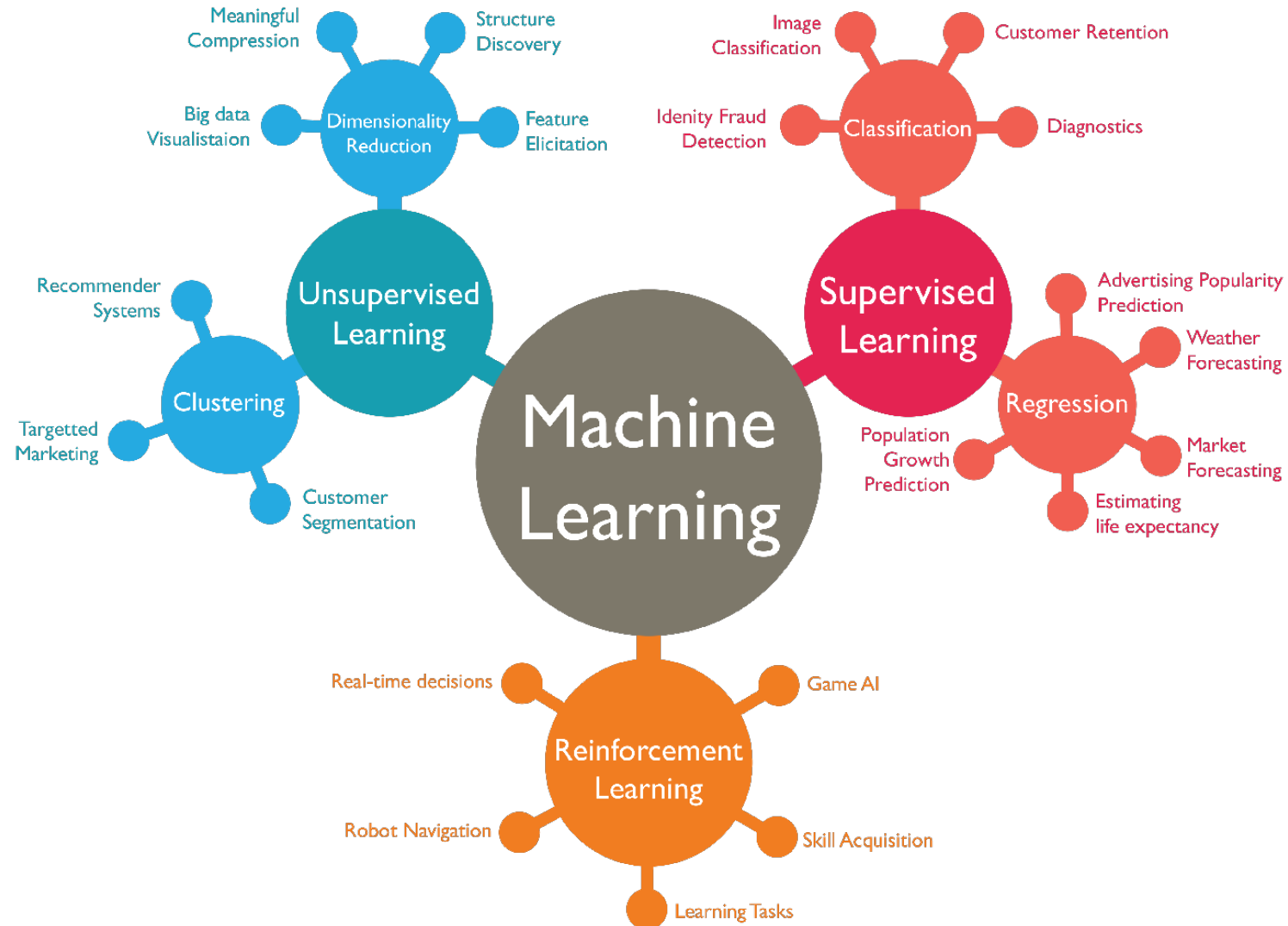




TYPE OF MACHINE LEARNING



TYPE OF MACHINE LEARNING



Machine learning models and their training algorithms

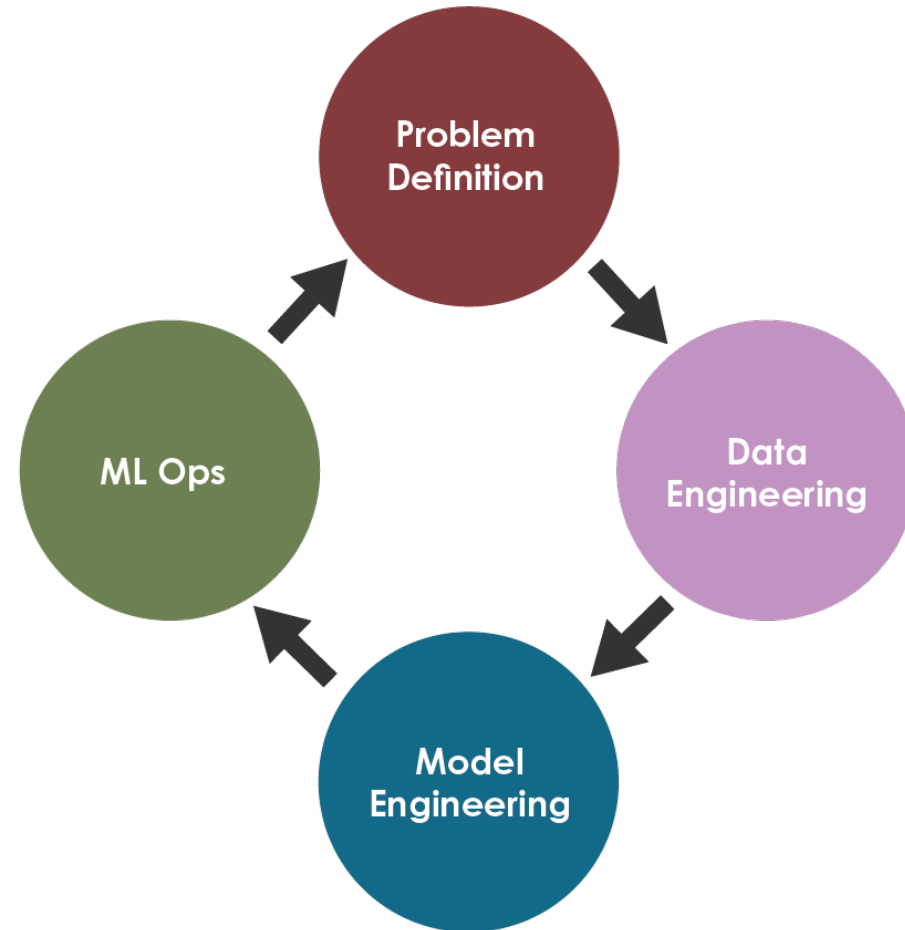
Supervised learning	Unsupervised learning	Semi-supervised learning	Reinforcement learning
Data scientists provide input, output and feedback to build model (as the definition). EXAMPLE ALGORITHMS: Linear regressions ■ Sales forecasting. ■ Risk assessment. Support vector machines ■ Image classification. ■ Financial performance comparison. Decision trees ■ Predictive analytics. ■ Pricing.	Use deep learning to arrive at conclusions and patterns through unlabeled training data. EXAMPLE ALGORITHMS: Apriori ■ Sales functions. ■ Word associations. ■ Searcher. K-means clustering ■ Performance monitoring. ■ Searcher intent. Artificial neural networks ■ Generate new, synthetic data. ■ Data mining and pattern recognition.	Builds a model through a mix of labeled and unlabeled data, a set of categories, suggestions and example labels. EXAMPLE ALGORITHMS: Generative adversarial networks ■ Audio and video manipulation. ■ Data creation. Self-trained Naïve Bayes classifier ■ Natural language processing.	Self-interpreting but based on a system of rewards and punishments learned through trial and error, seeking maximum reward. EXAMPLE ALGORITHMS: Q-learning ■ Policy creation. ■ Consumption reduction. Model-based value estimation ■ Linear tasks. ■ Estimating parameters.



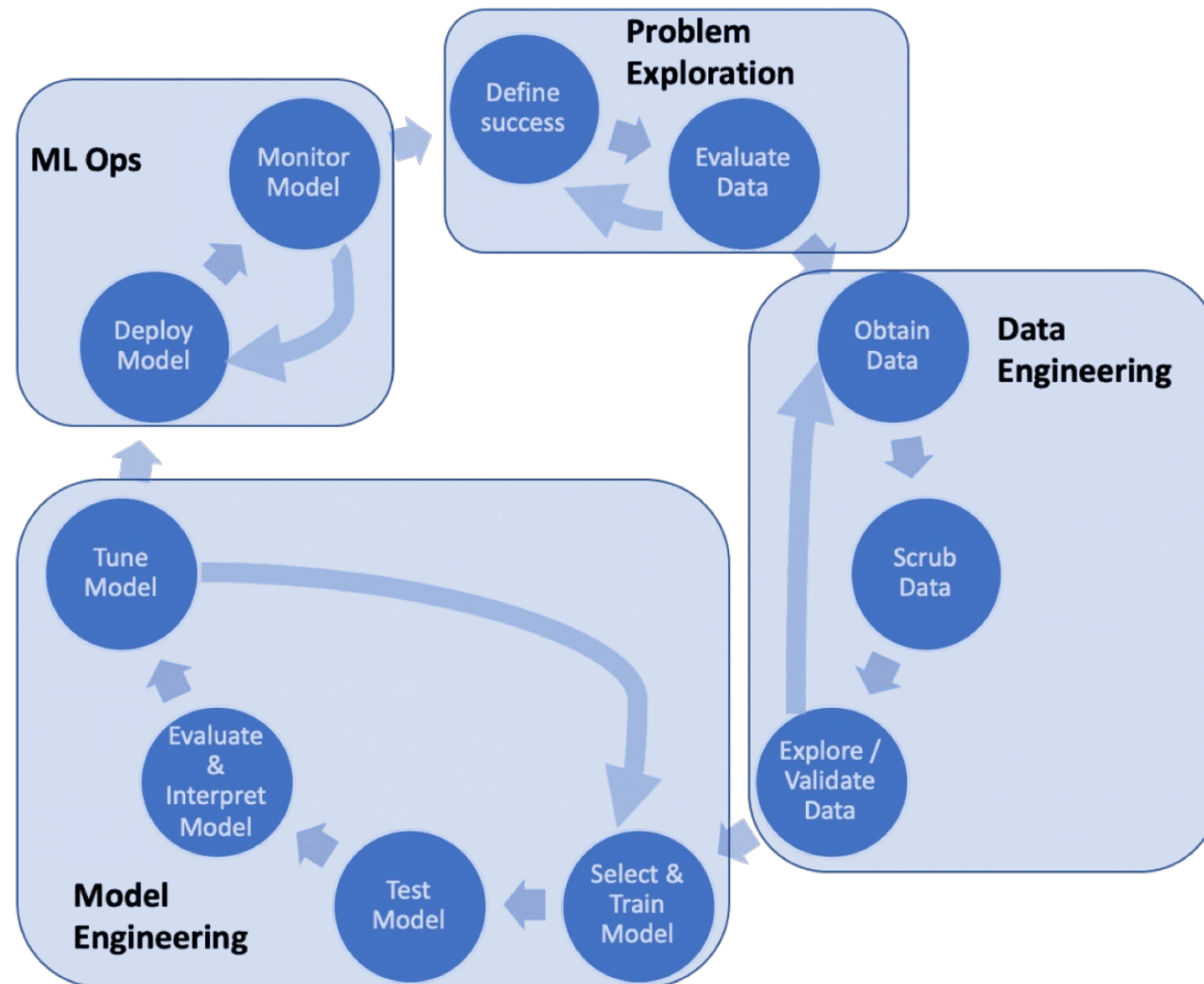
MACHINE LEARNING LIFE CYCLE



HIGH LEVEL ML PROCESS



DETAIL ML PROCESS



PROBLEM EXPLORATION

First focus on how the model will be used. In the process, assess the desired model accuracy and explore other details, such as if false positives are worse than false negatives. This phase also includes understanding what data might be available.

- **Define Success:** Define the problem to be solved. For example, what should be predicted. This helps define what data will be needed. Also, make sure it's clear how success will be measured.
- **Evaluate Data:** Determine what are the relevant data sources. In other words, evaluate what data the team will need, how that data is collected, and where the data is stored.



DATA ENGINEERING

Design and build data pipelines. These pipelines get, clean and transform data into a format that is more easily used to build a predictive model. Note that this data might be coming from multiple data sources, so merging the data is also a key aspect of data engineering. This is often where the most time is spent in an ML project.

- **Obtain Data:** Assembling the data. This includes connecting to remove data stored and databases, which might be in different formats. For example, some data might be in CSV format, and other data could be available in JSON via web services.
- **Scrub Data:** The process of re-formatting particular attributes and correcting errors in data, such as missing values imputation. Datasets are often missing values, or they may contain values of the wrong type or range. Cleaning can include removing duplicates, correcting errors, dealing with missing values, normalization, and handling data type conversions.
- **Explore / Validate Data:** Get a basic understanding of the data. This exploratory analysis includes data profiling to obtain information about the content and structure of the data. The goal is to both understand the data attributes as well as the quality of the data.



MODEL ENGINEERING

This is the phase that most people associate with building a machine learning model. During this phase, data is used to train and evaluate the model. This is often an iterative task, where the different models are tried, and the model is tuned.

- **Select & Train Model:** The process of identifying an appropriate model, and then building / training the model (on training data). The goal of training is to answer a question or make a prediction correctly as often as possible.
- **Test Model:** Run the model on data that the model has not yet seen (such as testing data). In other words, perform model testing by using data that was withheld from training (i.e., backtesting).
- **Evaluate & Interpret Model:** Objectively measure the performance of the model. Note that basic evaluation explores metrics such as accuracy and precision, to determine if the model is useable, and which model is best for the specific problem being explored. This evaluation also includes an understanding of when the model makes mistakes. More generally, validating the trained model helps to ensure the model meets original organizational objectives before the ML model is put into production.
- **Tune Model:** This step refers to parameter tuning, which, depending on the model being used, can be more an art than a science. In short, models typically have parameters (i.e., dials for tuning the model), which allows the model to get improved performance via parameter refinement. Simple model parameters may include attributes such as the number of training steps and the initialization of certain values.



ML OPS

Broadly defined, [machine learning operations](#) (ML Ops) spans a wide set of practices, systems, and responsibilities that data scientists, data engineers, cloud engineers, IT operations, and business stakeholders use to deploy, scale, and maintain machine learning solutions.

- **Deploy Model:** Package and put the model to use (i.e., into production). While this varies from one group to another, the team needs to understand the expected model performance, how the model will be monitored, and in general, key performance indicators (KPIs) of the model.
- **Monitor Model:** Maintain the model in production. This includes monitoring the KPIs and proactively working to ensure stable and robust predictions.



APPLICATION MACHINE LEARNING IN REAL-WORLD SCENARIO



HEALTHCARE

- **Disease Diagnosis:** ML models assist in diagnosing diseases such as cancer, diabetes, and heart conditions by analyzing medical images, genetic data, or patient records. For example, image recognition techniques help in identifying tumors from MRI scans.
- **Predictive Healthcare:** Predicting patient readmission rates, disease outbreaks, and personalized treatment plans based on patient history and current trends.



FINANCE

- **Fraud Detection:** ML algorithms analyze transaction patterns to detect suspicious activities, helping in real-time fraud prevention.
- **Algorithmic Trading:** Machine learning models predict stock price movements and make automated trades based on vast amounts of market data.
- **Credit Scoring:** Algorithms assess creditworthiness by analyzing customer financial history, making lending decisions more efficient and accurate.



RETAIL AND E-COMMERCE

- **Recommendation Systems:** E-commerce platforms like Amazon and Netflix use ML to recommend products and content based on user behavior and preferences.
- **Inventory Management:** ML models optimize stock levels, predicting demand based on past sales, seasons, and trends, reducing storage costs and wastage.
- **Customer Sentiment Analysis:** ML analyzes customer reviews and social media mentions to gauge customer satisfaction, allowing companies to improve their products and services.



TRANSPORTATION AND AUTONOMOUS VEHICLES

- **Self-Driving Cars:** Autonomous vehicles use ML to process data from sensors and cameras, enabling them to recognize obstacles, follow traffic rules, and navigate safely.
- **Traffic Prediction and Management:** ML algorithms forecast traffic patterns, allowing for optimized routes and congestion management, as seen in GPS navigation apps like Google Maps.
- **Predictive Maintenance:** ML helps in predicting when vehicles or machinery will need maintenance, reducing downtime and extending their life cycle.



MANUFACTURING AND SUPPLY CHAIN

- **Quality Control:** Computer vision models detect product defects on assembly lines, ensuring high-quality manufacturing standards.
- **Supply Chain Optimization:** ML algorithms forecast demand, optimize inventory, and streamline logistics, reducing costs and improving delivery times.
- **Robotics and Automation:** Machine learning is integrated into robots for tasks such as assembling parts, sorting, or inspecting products, making manufacturing more efficient.



NATURAL LANGUAGE PROCESSING (NLP)

- **Chatbots and Virtual Assistants:** ML models power conversational agents like Siri, Alexa, and customer support bots, which can understand and respond to queries in natural language.
- **Language Translation:** Services like Google Translate leverage ML to translate text or speech between languages, improving cross-cultural communication.
- **Sentiment Analysis:** ML tools analyze the sentiment of text, allowing businesses to gauge public opinion on products, brands, or services.



AGRICULTURE

- **Crop and Soil Monitoring:** ML models analyze soil health, crop yield patterns, and predict pest outbreaks, helping farmers optimize crop production.
- **Precision Farming:** By using data from sensors and satellite images, ML can guide efficient use of resources such as water, fertilizers, and pesticides.
- **Yield Prediction:** Machine learning models forecast crop yields based on weather, soil conditions, and historical yield data, aiding farmers in making informed decisions.



ENERGY AND UTILITIES

- **Energy Demand Forecasting:** Power companies use ML to predict energy consumption patterns, which helps in efficient power generation and distribution.
- **Smart Grid Management:** ML algorithms help balance electricity supply and demand in real time, reducing energy wastage and improving grid reliability.
- **Predictive Maintenance of Equipment:** ML monitors power plants and utility assets, detecting early signs of equipment failure to prevent outages.



ENTERTAINMENT AND SOCIAL MEDIA

- **Content Recommendations:** Platforms like YouTube, Spotify, and Instagram use ML to suggest videos, songs, and posts based on user preferences.
- **Content Creation and Moderation:** Machine learning can assist in creating captions, detecting inappropriate content, and managing user comments.
- **Audience Engagement Analysis:** ML algorithms help analyze user behavior, providing insights on what content users engage with most, enabling better content strategies.

